

Ind-classification

```
library(MASS)
library(mixlm)

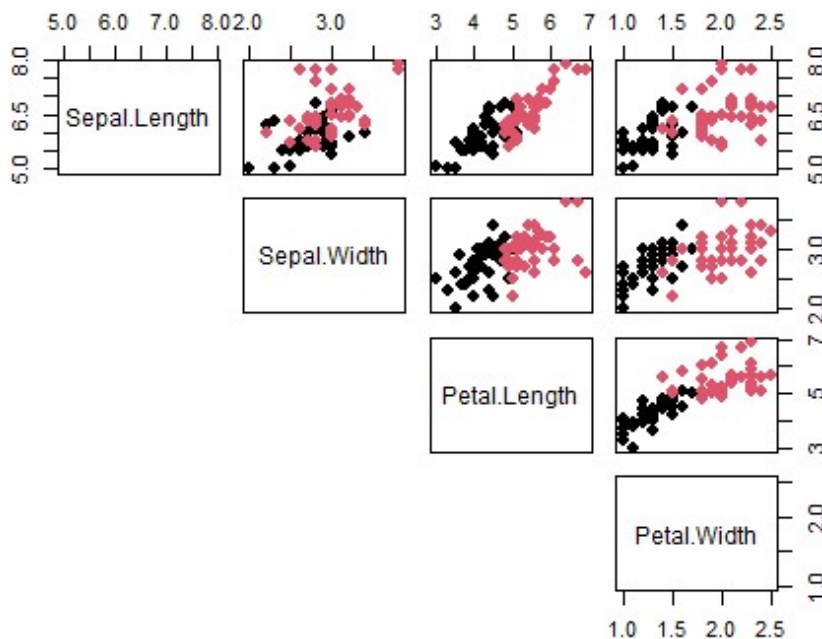
##
## Attaching package: 'mixlm'

## The following objects are masked from 'package:stats':
##
##      glm, lm

#load dataset

load("iris.train.Rdata")
load("iris.test.Rdata")

#a
```



#b Discrimination corresponds to the model fitting process in statistical inference. We seek good variables for discriminating classes. In other words, we discriminate by using existing observations to find the variables that best separate these observations in their known groups or classes. Classification corresponds to the actual prediction of new samples, that is, to allocate new samples to classes using the presumably best “classifier”-model estimated from the training data. In other words, we classify when predicting the

corresponding (unknown) group or class for a new or several new observations. Then, to classify, we need to know in advance the best variables that will allow us to find the best model that we found first, using discrimination. #c This means that both species are equally probable if you sample a random plant from the population. #d

```
mod1.lda <- lda(Species ~ Sepal.Length, data = iris.train, prior = c(0.5,
0.5))
mod1.lda
```

```
## Call:
## lda(Species ~ Sepal.Length, data = iris.train, prior = c(0.5,
##      0.5))
##
## Prior probabilities of groups:
## versicolor  virginica
##      0.5      0.5
##
## Group means:
##      Sepal.Length
## versicolor      5.8950
## virginica      6.5925
##
## Coefficients of linear discriminants:
##      LD1
## Sepal.Length 1.885221
```

#e

```
mod1.lda.cv <- lda(Species ~ Sepal.Width, data = iris.train, prior = c(0.5,
0.5), CV = TRUE)
confusion(iris.train$Species, mod1.lda.cv$class)
```

```
##      True
## Predicted versicolor virginica
## versicolor      22      16
## virginica      18      24
## Total          40      40
## Correct        22      24
##
## Proportions correct
## versicolor  virginica
##      0.55      0.60
##
## N correct/N total = 46/80 = 0.575
```

The confusion matrix is given in the first part of the output. 32 out of 40 versicolor are correctly classified, whereas 29 out of 40 virginica are correct. In total 61 out of 80 are correctly classified giving an accuracy of 0.7625 as reported. The APER is 1-accuracy, hence $APER = 1 - 0.7625 = 0.2375$. We typically seek classifiers that minimize the classification error rate.

#f

```
mymod <- qda(Species ~ Petal.Length + Petal.Width, data = iris.train, prior =
c(0.5, 0.5))
mymod.cv <- qda(Species ~ Petal.Length + Petal.Width, data = iris.train,
prior = c(0.5, 0.5), CV = TRUE)
confusion(iris.train$Species, mymod.cv$class)

##              True
## Predicted    versicolor virginica
## versicolor      38         2
## virginica        2        38
## Total           40        40
## Correct          38        38
##
## Proportions correct
## versicolor  virginica
##      0.95      0.95
##
## N correct/N total = 76/80 = 0.95
```

This has an accuracy of 0.95 and, hence, an error of 0.05.

#g

In LDA we assume equal variance structure for all classes, whereas in QDA we assume different variance structures for all classes.

#h

```
pred <- predict(mymod, newdata = iris.test) #use the QDA model to classify
new observations from new data set "iris.test"
confusion(iris.test$Species, pred$class) #checking accuracy of the model for
the new dataset

##              True
## Predicted    versicolor virginica
## versicolor      10         0
## virginica        0        10
## Total           10        10
## Correct          10        10
##
## Proportions correct
## versicolor  virginica
##      1      1
##
## N correct/N total = 20/20 = 1
```

The model I found in gave a perfect classification for the new data, accuracy=1.0 and error=0.0.

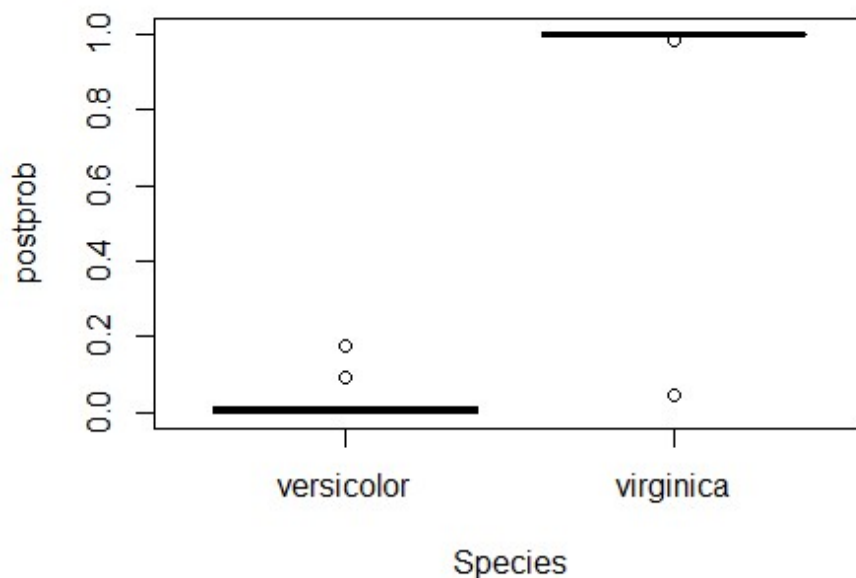
#i (optional) Classification using logistic regression:

```

mymod2 <- glm(Species ~ Petal.Length + Petal.Width, family = binomial, data =
iris.train)
iris.test$postprob <- predict(mymod2, new = iris.test, type = "response")
#argument type = "response" gives the predicted probabilities, which in this
case are the posterior probabilities of the observations

```

Using the posterior probabilities predicted from this model, we will then classify the the observations from iris.test dataset.



The boxplot shows that most of the “virginica” observations have the posterior probabilities more than 0.5 so we will classify the observations with $\text{postprob} > 0.5$ as virginica:

```

predicted <- factor(ifelse(iris.test$postprob >
0.5, "virginica", "versicolor")) #classify the observations with postprob > 0.5
as "virginica"
confusion(iris.test$Species, predicted) #make the confusion matrix

##           True
## Predicted  versicolor virginica
## versicolor      10         1
## virginica       0         9
## Total          10        10
## Correct         10         9
##
## Proportions correct
## versicolor  virginica
##          1.0         0.9

```

```
##
```

```
## N correct/N total = 19/20 = 0.95
```